

PEARSON EDEXCEL INTERNATIONAL AS/A LEVEL

BIOLOGY (IAL)

Complete Core Practicals Revision Notes

Unit 6 Examination — All 18 Core Practicals (AS + A2)

Covers: IAS Units 1–3 (CPs 1–9) | IA2 Units 4–5 (CPs 10–18) | Full Lab Skills

CORE PRACTICALS OVERVIEW — ALL 18

#	Core Practical Title	Unit
CP1	Semi-quantitative methods: Benedict's reagent (reducing sugars) & iodine (starch)	IAS Unit 1
CP2	Investigate vitamin C content of food and drink	IAS Unit 1
CP3	Investigate membrane properties: effect of alcohol & temperature on membrane permeability	IAS Unit 2
CP4	Investigate effects of temperature, pH, enzyme conc. & substrate conc. on initial rate of enzyme reactions	IAS Unit 2
CP5	Use a light microscope to observe/draw animal cells; use graticule to measure (scale)	IAS Unit 2
CP6	Prepare and stain a root tip squash to observe stages of mitosis	IAS Unit 2
CP7	Light microscope: observe TS roots, stems, leaves; identify sclerenchyma, phloem, xylem vessels	IAS Unit 3
CP8	Determine the tensile strength of plant fibres	IAS Unit 3
CP9	Investigate antimicrobial properties of plants; aseptic techniques for safe handling of bacteria	IAS Unit 3
CP10	Investigate effects of light intensity, wavelength, temperature & CO ₂ availability on rate of photosynthesis	IA2 Unit 4
CP11	Ecology study: quadrats & transects for distribution/abundance; measure abiotic factors	IA2 Unit 4
CP12	Investigate effects of temperature on development of organisms	IA2 Unit 4
CP13	Investigate rate of growth of microorganisms in liquid culture	IA2 Unit 4
CP14	Investigate effect of different antibiotics on bacteria	IA2 Unit 4
CP15	Use artificial hydrogen carrier (redox indicator) to investigate respiration in yeast	IA2 Unit 5
CP16	Use a simple respirometer to determine rate of respiration & RQ of a material	IA2 Unit 5
CP17	Investigate effects of exercise on tidal volume, breathing rate, RMV & O ₂ consumption from spirometer traces	IA2 Unit 5
CP18	Investigate the production of amylase in germinating cereal grains	IA2 Unit 5

CORE PRACTICAL 1 (IAS Unit 1)

Semi-Quantitative Methods: Benedict's Reagent (Reducing Sugars) & Iodine Solution (Starch) Using Colour Standards

SPECIFICATION REFERENCE

1.3 — Food molecules: Carbohydrates

Background & Principle

This practical uses colorimetric comparison techniques to estimate concentrations of reducing sugars and starch in an unknown solution by matching colour against prepared standards.

- Benedict's test: Reducing sugars (e.g. glucose, maltose) reduce Cu^{2+} ions in alkaline solution $\rightarrow$ insoluble Cu_2O precipitate. Colour ranges: blue (0) $\rightarrow$ green $\rightarrow$ yellow $\rightarrow$ orange $\rightarrow$ brick-red (high conc.)
- Iodine test: Iodine (I_2 /KI) interacts with starch's helical amylose structure $\rightarrow$ blue-black colour. Intensity increases with starch concentration.

Part A: Reducing Sugars — Benedict's Reagent

Apparatus & Reagents

- Stock 2% glucose solution, distilled water, syringes (1 cm^3 , 5 cm^3)
- Benedict's reagent, water bath at 80–95 $^\circ\text{C}$ (or boiling water bath)
- Labelled test tubes ($\times 6$), test tube rack, colourimetric chart

Method (Step-by-step)

- Prepare a serial dilution: use the 2% stock glucose to make colour standards of 2%, 1%, 0.5%, 0.2%, 0.1%, and 0% (distilled water) — each 5 cm^3 in labelled test tubes.
- Prepare a separate test tube labelled 'unknown' with 5 cm^3 of the unknown fruit juice / food sample.
- Add 2 cm^3 of Benedict's reagent to EACH test tube using a clean syringe.
- Place all test tubes in a water bath at 80 $^\circ\text{C}$ for 5 minutes (or boil for 2 minutes).
- Remove and allow to cool; compare the colour of the unknown tube to the colour standards.
- Record the closest match — this gives a semi-quantitative estimate of reducing sugar concentration.

Results & Analysis

Colour Observed	Approximate Reducing Sugar Concentration
Blue (no change)	0% (absent)
Green	~0.1%
Yellow	~0.5%
Orange	~1%
Brick-red precipitate	~2% or above

Part B: Starch — Iodine Solution

Apparatus & Reagents

- Stock 2% starch solution, iodine solution (I_2 /KI), distilled water
- Syringes, 6 labelled test tubes, white tile (for colour comparison)

Method

7. Prepare starch colour standards: 2%, 1%, 0.5%, 0.2%, 0.1%, and 0% (water blank).
8. Add 0.5 cm³ of iodine solution to each test tube.
9. Label a 6th tube 'unknown' and add 0.5 cm³ iodine to the unknown starch solution.
10. Compare the colour of the unknown against the standards immediately (iodine colour is transient at high temperatures).

KEY RISKS & SAFETY
• Benedict's reagent: irritant — avoid skin contact; hot water bath hazard → use tongs
• Iodine solution: irritant, stains skin — wear gloves and eye protection
• Boiling water: burn risk — use tongs; allow test tubes to cool before handling

Evaluation & Sources of Error

- Colour matching is subjective — use a colorimeter for more precise measurement
- Ensure same volume of reagent and sample are used for all standards and unknown
- Benedict's: temperature and time must be identical for all tubes; variation will affect colour intensity
- Iodine: read results immediately as iodine can evaporate; temperature affects result — keep at room temperature

CORE PRACTICAL 2 (IAS Unit 1)

Investigate the Vitamin C Content of Food and Drink

SPECIFICATION REFERENCE

1.3 — Biochemical tests; Vitamin C (ascorbic acid)

Background & Principle

DCPIP (dichlorophenolindophenol) is a blue-coloured redox dye that is DECOLOURISED (reduced) by vitamin C (ascorbic acid). The volume of juice/food solution required to decolourise a fixed volume of DCPIP is inversely proportional to its vitamin C concentration.

- More vitamin C → less volume needed → higher concentration
- Used as a titration-style (volumetric) method

Apparatus & Reagents

- 0.1% DCPIP solution, fresh fruit juices (e.g. orange, lemon), vitamin C standard solution (1 mg/cm³)
- Burette or syringe/graduated pipette, white tile, conical flask/boiling tube
- Balance, measuring cylinders

Method

11. Using a syringe/pipette, transfer exactly 1 cm³ of DCPIP into a boiling tube or small conical flask.
12. Using a graduated burette or syringe, add the vitamin C standard solution (1 mg/cm³) drop by drop to the DCPIP, swirling after each drop.
13. Record the volume of standard solution needed to just decolourise (turn from blue to colourless) the DCPIP. This is the titre.
14. Repeat steps 1–3 with each fruit juice or food extract instead of the standard.
15. Calculate the vitamin C concentration: $\text{Conc} = (\text{volume of standard} \times \text{conc of standard}) \div \text{volume of juice used}$.

CALCULATING VITAMIN C CONCENTRATION

Volume of standard to decolourise DCPIP = V_{std} cm³ (at 1 mg/cm³)

Volume of juice to decolourise DCPIP = V_{juice} cm³

Vitamin C in juice = $(V_{\text{std}} \times 1 \text{ mg/cm}^3) / V_{\text{juice}} = \text{mg/cm}^3$

Example: $V_{\text{std}} = 0.2 \text{ cm}^3$; $V_{\text{juice}} = 0.5 \text{ cm}^3$

Vit C = $(0.2 \times 1) / 0.5 = 0.4 \text{ mg/cm}^3$

Controls

- Use distilled water as a negative control (should not decolourise DCPIP)
- Repeat at least 3 times for each sample → calculate mean titre

SAFETY

• DCPIP: irritant — wear gloves and eye protection
• Vitamin C: generally safe but use lab-grade solutions carefully

Sources of Error & Improvements

Source of Error	Improvement
Parallax error reading burette/syringe	Read at eye level; use a white background
Endpoint subjective (colour change faint)	Use a colorimeter set at 600 nm for precision
Vitamin C oxidises over time (in air)	Prepare fresh samples; perform experiment quickly
Juice turbidity interferes with colour	Filter juice before use

CORE PRACTICAL 3 (IAS Unit 2)

Investigate Membrane Properties: Effect of Alcohol and Temperature on Membrane Permeability

SPECIFICATION REFERENCE

2.2 — Cell membranes; fluid mosaic model; factors affecting permeability

Background & Principle

Beetroot cells contain red/purple pigment (betacyanin) in their vacuoles, enclosed by the tonoplast and cell surface membrane. Both alcohol and high temperature damage the phospholipid bilayer, increasing permeability and causing pigment leakage.

- Temperature: denaturing proteins in the membrane; increased phospholipid fluidity → more permeable
- Alcohol: dissolves phospholipids (lipid-soluble); disrupts the bilayer structure

Leakage is measured using a colorimeter (absorbance at ~540 nm for red/purple) or by visual comparison.

PART A: Effect of Temperature

Variables

Variable Type	Detail
Independent	Temperature (e.g. 20, 30, 40, 50, 60, 70, 80 °C)
Dependent	Absorbance of solution at 540 nm (colorimeter)
Controlled	Size of beetroot cylinders, volume of water, time in water bath, pH

Method

16. Cut beetroot into uniform cylinders (e.g. 3 cm length, using a cork borer for consistent diameter).
17. Rinse cylinders thoroughly in cold distilled water until no more colour leaks — this removes pigment released during cutting.
18. Set up water baths at a range of temperatures (e.g. 20, 30, 40, 50, 60, 70, 80 °C).
19. Place one beetroot cylinder in 5 cm³ of distilled water in a test tube at each temperature for exactly 15 minutes.
20. Remove the beetroot and allow solutions to reach room temperature.
21. Zero (blank) the colorimeter with distilled water, then measure the absorbance of each solution at 540 nm.
22. Repeat each temperature at least 3 times and calculate mean absorbance.

PART B: Effect of Alcohol Concentration

23. Prepare a dilution series of ethanol: e.g. 0%, 5%, 10%, 20%, 40%, 60%, 80%.
24. Place rinsed, uniform beetroot cylinders into 5 cm³ of each ethanol concentration for 15 min at room temperature.
25. Remove beetroot; measure absorbance of each solution at 540 nm.

Expected Results

- Temperature: absorbance increases above ~40 °C and rises steeply above 60 °C (denaturing membrane proteins)
- Alcohol: absorbance increases with increasing alcohol concentration (disruption of phospholipid bilayer)

COLORIMETER TECHNIQUE

- Calibrate (zero) with distilled water blank before measuring any samples
- Use filter: green/blue (for red/purple pigment — complementary colour absorbs most)
- Wipe cuvette exterior; ensure no air bubbles in solution
- Higher absorbance = more pigment leaked = more permeable membrane

SAFETY

- Ethanol: highly flammable — keep away from flame; work in ventilated area
- Hot water baths: burn risk — use tongs; do not use >90 °C
- Beetroot: stains skin and clothing — wear gloves and apron

Analysis & Evaluation

Source of Error	Improvement
Unequal beetroot cylinder size	Use cork borer + ruler; measure diameter & length
Not enough rinsing before experiment	Rinse until water runs clear
Temperature fluctuations	Use thermostatically controlled water baths
Colorimeter reading variation	Use same colorimeter; blank each time; repeat × 3

CORE PRACTICAL 4 (IAS Unit 2)

Investigate the Effects of Temperature, pH, Enzyme Concentration and Substrate Concentration on the Initial Rate of Enzyme-Catalysed Reactions

SPECIFICATION REFERENCE

2.2 — Enzymes: factors affecting enzyme activity; Michaelis-Menten kinetics

Background & Principle

Enzymes are biological catalysts. Their activity is influenced by temperature, pH, enzyme concentration, and substrate concentration. Initial rate is measured because it represents conditions before substrate depletion or product inhibition.

- Trypsin/protease + casein (milk substrate): as casein is hydrolysed, milk becomes transparent — measured by colorimeter or turbidimeter
- Alternatively: catalase + H_2O_2 (measure O_2 produced), amylase + starch (iodine sampling method)

Using Trypsin & Casein (Recommended Method)

Variables Summary

Factor Investigated	IV Range	Dependant Variable
Temperature	10, 20, 30, 40, 50, 60, 70 °C	Absorbance at 550 nm (time to clear)
pH	3, 4, 5, 6, 7, 8, 9, 10	Absorbance or time to reach set absorbance
Enzyme conc.	0.1%, 0.25%, 0.5%, 1%, 2%	Absorbance change per minute
Substrate conc.	0.1%, 0.25%, 0.5%, 1%, 2%	Absorbance change per minute

Method (Temperature Variation)

26. Prepare sets of test tubes: one set containing 1% casein solution (5 cm³ each), one set containing 0.1% trypsin (2 cm³ each).
27. Place matched pairs into water baths at each temperature for 5 minutes to equilibrate.
28. Mix trypsin into each casein tube (total 7 cm³). Start stopwatch immediately.
29. Take absorbance readings at 550 nm every 1–2 minutes using the colorimeter.
30. Calculate initial rate: gradient of the absorbance vs. time graph in the linear (initial) region.
31. Plot initial rate vs. temperature; identify the optimum temperature.

Method (pH Variation)

32. Prepare buffer solutions at different pH values (e.g. citrate buffer for pH 3–6, phosphate buffer for pH 6–8, carbonate buffer for pH 9–10).
33. Add equal volumes of buffer to trypsin solutions to establish pH; equilibrate.
34. Mix trypsin + casein + buffer and measure absorbance over time as above.

Calculating Initial Rate

INITIAL RATE CALCULATION

Plot absorbance (y-axis) against time (x-axis)
Draw best-fit curve
Initial rate = gradient of tangent drawn at $t = 0$
OR if linear at start: Initial rate = $\Delta\text{Absorbance} / \Delta\text{time}$ (in the first linear section)
Units: absorbance units per minute (AU min^{-1}) or AU s^{-1}

Effect of Substrate Concentration (Michaelis-Menten)

- At low $[S]$: rate is proportional to $[S]$ (enzyme not saturated)
- At high $[S]$: rate plateaus (V_{max}) — all enzyme active sites occupied
- $K_m = [S]$ at half-maximal rate — indicates enzyme affinity for substrate

SAFETY
• Trypsin: irritant — wear gloves
• pH buffers: some are corrosive (strong acid/base) — check COSHH
• Hydrogen peroxide (if used instead): irritant/oxidiser — avoid skin/eye contact

Evaluation

Source of Error	Improvement
Mixing affects 'time zero'	Add enzyme to substrate consistently; start timing at point of mixing
Temperature fluctuations during assay	Use thermostatically controlled water baths throughout
Enzyme denatures at extremes	Note this in analysis; it explains drop-off at high temp/extreme pH
Buffer inaccuracy at given pH	Use pH meter to verify buffer pH before use

CORE PRACTICAL 5 (IAS Unit 2)

(i) Use a Light Microscope to Observe and Draw Animal Cells; (ii) Use a Graticule to Make Measurements and Understand Scale

SPECIFICATION REFERENCE

2.1 — Cell structure; microscopy; scale

Background

Light microscopy allows observation of cells at magnifications up to $\sim \times 1500$. Understanding scale and making accurate measurements are essential practical skills in biology.

Part (i): Observing & Drawing Animal Cells

Suitable Materials

- Human cheek cells (buccal epithelium) — most common
- Red blood cells (human or frog erythrocytes) — for comparison

Method: Cheek Cell Preparation

35. Gently scrape the inside of your cheek with a clean, flat toothpick or cotton swab.
36. Smear the sample onto a clean, grease-free glass slide.
37. Add one drop of methylene blue stain (or crystal violet) to improve contrast.
38. Lower a coverslip at an angle to avoid air bubbles.
39. Observe under the microscope: start with low power objective ($\times 4$), then move to $\times 10$, then $\times 40$ for detail.
40. Focus carefully using coarse then fine adjustment.
41. Make a large, clear, labelled scientific drawing.

Rules for Biological Drawing

- Use a sharp pencil; draw in continuous lines (no shading, no sketching)
- Drawings should be large (at least half an A4 page for single cell)
- Label lines should be horizontal, drawn with a ruler, and NOT crossing each other
- Magnification should be stated: e.g. $\times 400$
- Nucleus, cell membrane, cytoplasm should all be clearly labelled in cheek cells

Part (ii): Using an Eyepiece Graticule (Scale)

An eyepiece graticule (micrometer) is an internal scale within the eyepiece. It must be calibrated using a stage micrometer (a glass slide with a scale marked in 0.1 mm and 0.01 mm divisions).

Calibration Procedure

42. Place the stage micrometer on the stage and bring it into focus.
43. Align the eyepiece graticule scale with the stage micrometer scale.
44. Count how many eyepiece graticule divisions (e.g. 40 EPD) correspond to a known stage micrometer distance (e.g. $200\text{ }\mu\text{m} = 0.2\text{ mm}$).
45. Calculate: $1\text{ EPD} = 200\text{ }\mu\text{m} \div 40 = 5\text{ }\mu\text{m}$.
46. This calibration value applies ONLY for that specific objective lens — recalibrate for each magnification.

CALIBRATION CALCULATION
Stage micrometer: 1 small division = 10 μm (0.01 mm)
Example: 50 EPD aligns with 30 stage divisions = $30 \times 10 \mu\text{m} = 300 \mu\text{m}$
$\rightarrow 1 \text{ EPD} = 300 / 50 = 6 \mu\text{m}$
To measure a cell: count EPD across the cell $\times$ calibration factor
e.g. Cell = 8 EPD wide $\rightarrow 8 \times 6 = 48 \mu\text{m}$

Calculating Magnification

MAGNIFICATION FORMULA
Magnification = Image size / Actual size
Rearranged: Actual size = Image size / Magnification
Image size = Actual size $\times$ Magnification
IMPORTANT: Use consistent units (both in μm , mm, or cm)
Convert: 1 mm = 1000 μm ; 1 cm = 10 mm = 10,000 μm

SAFETY
• Methylene blue / crystal violet: skin/clothing stain — wear gloves
• Glass slides: sharp edges — handle carefully; dispose in sharps bin
• Cheek cells: use your own cheek only; discard swabs safely (biological waste)

CORE PRACTICAL 6 (IAS Unit 2)

Prepare and Stain a Root Tip Squash to Observe the Stages of Mitosis

SPECIFICATION REFERENCE

2.3 — Mitosis; cell cycle; chromosome behaviour

Background

Root tips are ideal for observing mitosis because cells in the meristem (region of cell division) are actively dividing. The cell cycle has 4 main stages: Interphase, Prophase, Metaphase, Anaphase, Telophase. Squashing spreads cells in a single layer for clear observation.

Apparatus & Materials

- Garlic bulb or onion bulb (grown in water for 2–3 days to produce 1–2 cm roots)
- 1 mol dm⁻³ hydrochloric acid (HCl), acetocarmine or aceto-orcein stain
- Water baths at 55 °C and 60 °C, watch glass, mounted needle, forceps
- Glass slides, coverslips, thumb (for squashing), light microscope

Method

47. Cut off approximately 5 mm from the tip of a root (pale yellow growing tip, not the white older region).
48. Place the root tip in a watch glass with 1 mol dm⁻³ HCl that has been pre-warmed to 55 °C in a water bath.
49. Leave in the acid for exactly 5 minutes at 55 °C — this hydrolyses (softens) the middle lamella between cells.
50. Remove the root tip and blot dry on filter paper. Transfer to a clean slide.
51. Add 1–2 drops of acetocarmine or aceto-orcein stain. Leave for 2 minutes.
52. Use a mounted needle to tease apart the root tip into a small fragment of cells.
53. Lower a coverslip over the cells. Place a piece of folded filter paper on top.
54. Using your thumb, press firmly downward (do NOT twist) on the coverslip to squash cells into a single layer.
55. Observe under low power (×40), then switch to high power (×100 or ×400) to identify mitotic stages.

Identifying Mitotic Stages

Stage	Key Features Observed	Chromosome Behaviour
Interphase	Large nucleus visible; nucleolus present; no distinct chromosomes	Chromosomes uncoiled; DNA replication occurs (S-phase)
Prophase	Chromosomes condense and become visible; nucleolus disappears	Chromosomes coil; centrioles (if present) move to poles
Metaphase	Chromosomes align at equator (cell plate); spindle fully formed	Chromosomes attached to spindle fibres at centromeres
Anaphase	Chromosomes being pulled apart to opposite poles; cell elongates	Sister chromatids separate; move to poles
Telophase	Two sets of chromosomes at poles; nuclear envelope reforms	Chromosomes uncoil; cytokinesis begins

Calculating Mitotic Index

MITOTIC INDEX FORMULA
Mitotic Index = (number of cells in mitosis) / (total number of cells observed)
Multiply by 100 for percentage
e.g. 15 cells in mitosis out of 200 total → MI = $15/200 = 0.075$ (or 7.5%)
Higher MI = faster rate of cell division (e.g. cancer cells vs normal cells)

The relative time spent in each stage can be estimated from the proportion of cells in each stage — more cells in a stage means more time is spent in that stage.

SAFETY
• 1 mol dm ⁻³ HCl: corrosive — wear gloves and eye protection; do not inhale
• Acetocarmine/aceto-orcein: contains glacial acetic acid — corrosive; may be carcinogenic — handle with care, in fume cupboard if possible
• Hot water bath (55 °C): burn risk — use tongs
• Razor blades for cutting roots: sharp — cut away from body

CORE PRACTICAL 7 (IAS Unit 3)

Light Microscope: Observe (i) TS of Roots, Stems, Leaves; (ii) Plant Tissues; (iii) Identify Sclerenchyma Fibres, Phloem Sieve Tubes, Xylem Vessels and Their Location

SPECIFICATION REFERENCE

3.2 — Plant tissues: xylem, phloem, sclerenchyma; transport in plants

Background

Plant organs are composed of specialised tissues. In transverse sections (TS), these tissues appear in characteristic arrangements. Being able to identify them under the microscope is essential.

Pre-made Slides: What to Observe

Permanent stained slides of pre-made TS sections are used. Safranin stains lignified cell walls (xylem, sclerenchyma) red/orange; toluidine blue stains phloem blue.

(i) Transverse Section of a Dicot Root

Tissue / Feature	Location & Appearance
Epidermis	Outer single cell layer; may have root hair cells
Cortex	Large, loosely packed cells (parenchyma); stores starch
Endodermis	Inner boundary of cortex; Casparian strip visible in some stains
Pericycle	One cell layer inside endodermis; gives rise to lateral roots
Xylem	CENTRAL — star-shaped (e.g. 4-armed = tetrarch); thick red walls; no living contents
Phloem	Between xylem arms; sieve tube elements + companion cells; thinner walls

(ii) Transverse Section of a Dicot Stem

Tissue / Feature	Location & Appearance
Epidermis	Outer layer; may have cuticle; stomata not present in stem TS
Cortex	Between epidermis and vascular bundles; parenchyma + collenchyma at corners
Vascular bundles	Arranged in a RING; each bundle = xylem (inner) + phloem (outer) + cambium
Xylem	Large, thick-walled vessels; inner part of each bundle; lignified (red with safranin)

Phloem	Outer part of bundle; smaller cells; sieve tubes visible
Pith	Central parenchyma cells; large, thin-walled
Sclerenchyma fibres	Often form a 'cap' on outside of phloem; very thick walls; no living contents

(iii) Transverse Section of a Leaf (Dicot)

Tissue / Feature	Location & Appearance
Upper epidermis	Transparent cells; may have cuticle; no chloroplasts
Palisade mesophyll	Long, tightly packed cells just below upper epidermis; many chloroplasts
Spongy mesophyll	Loosely arranged cells; many air spaces; fewer chloroplasts
Lower epidermis	Contains stomata (guard cells); thinner cuticle
Vascular bundle (midrib)	Central region; xylem above phloem; surrounded by bundle sheath
Guard cells	Kidney-shaped; control stomatal aperture; contain chloroplasts

Identifying Key Cell Types

Xylem Vessels

- Dead cells; thick, lignified walls (appear red/dark with safranin); hollow lumen; no nucleus or cytoplasm
- Function: water and mineral transport; structural support

Phloem Sieve Tube Elements

- Living cells; thin walls; sieve plates at ends (perforated); no nucleus at maturity
- Companion cells: small cells with large nucleus alongside sieve tubes (metabolic support)
- Function: transport of organic solutes (sucrose, amino acids) by mass flow

Sclerenchyma Fibres

- Dead cells; uniformly very thick secondary cell walls (lignified); no lumen visible
- Function: mechanical support only

Drawing from Microscope Slides

Draw a plan diagram (showing tissue regions, NOT individual cells) and a detail diagram (showing individual cells at high power). Label all tissues.

SAFETY
• Glass slides: handle carefully; dispose in sharps bin
• Staining solutions (safranin, toluidine blue): irritant; stain skin/clothing

CORE PRACTICAL 8 (IAS Unit 3)

Determine the Tensile Strength of Plant Fibres

SPECIFICATION REFERENCE

3.2 — Sclerenchyma fibres; plant support; cellulose

Background

Tensile strength is the maximum force a material can withstand before breaking under tension (pulling force). Sclerenchyma fibres (e.g. from ramie, hemp, or leaf midrib) resist stretching due to their thick lignified cell walls. This practical investigates what factors affect fibre tensile strength.

Apparatus

- Plant fibres (e.g. from ramie, celery, lily leaf midrib, or similar)
- Clamp stand, G-clamp, metre ruler, masses (50 g, 100 g, etc.), mass holder
- Tape or adhesive for securing fibre ends, ruler for measuring fibre length & diameter

Method

56. Prepare plant fibres of known length (e.g. 10 cm) and, if possible, similar diameter. Record initial length.
57. Clamp the top end of the fibre securely to a stand.
58. Attach a mass hanger to the lower end.
59. Add masses in 50 g increments (or as appropriate); wait 30 seconds after each addition.
60. Record the mass at which the fibre breaks. Calculate force ($F = \text{mass} \times g$; $g = 9.81 \text{ m/s}^2$).
61. Repeat for multiple fibres (minimum 5) and calculate mean breaking force.
62. Investigate variables: compare dry vs. soaked fibres, fibres from different species, different diameters.

Calculating Tensile Strength

TENSILE STRENGTH FORMULA
Tensile Strength = Breaking Force (N) / Cross-sectional Area (m^2 or mm^2)
Cross-sectional area = $\pi \times r^2$ (if fibre assumed to be circular)
where r = radius of fibre (measured with a micrometer or ruler)
Units: N m^{-2} (Pascals) or N mm^{-2}

Variables to Investigate

Variable	Expected Effect
Soaking in water	May reduce strength (disrupts hydrogen bonds in cellulose)
Species of plant	Different lignification/cellulose content → different tensile strength

Fibre diameter	Larger diameter → greater breaking force; calculate tensile strength to compare fairly
Length of fibre	Length alone should not affect tensile strength if cross-section is same

SAFETY

- Falling masses: keep feet away from below the apparatus
- Fibre snapping under tension: wear eye protection; stand to the side
- Sharp blades (if cutting fibres): cut away from body

CORE PRACTICAL 9 (IAS Unit 3)

Investigate the Antimicrobial Properties of Plants, Including Aseptic Techniques for the Safe Handling of Bacteria

SPECIFICATION REFERENCE

3.3 — Immunity; antimicrobial substances; plants vs pathogens

Background

Many plants produce secondary metabolites with antimicrobial properties (e.g. allicin from garlic, terpenes, phenolics). These inhibit bacterial growth. Aseptic technique prevents contamination of cultures and the environment.

Aseptic Technique — Key Principles

ASEPTIC TECHNIQUE RULES

1. Work near a lit Bunsen burner or in a laminar flow cabinet (updraft prevents settlement of microbes)
2. Flame the neck of bottles/tubes before and after opening
3. NEVER leave agar plates open longer than necessary
4. Sterilise inoculation loops by flaming to red-hot before and after use
5. Label plates on the base (bottom), not the lid
6. Incubate plates INVERTED (upside down) to prevent condensation dripping onto colonies
7. Incubate at $\leq 30^{\circ}\text{C}$ for school practicals (not 37°C which encourages human pathogens)
8. Autoclave or disinfect all waste materials before disposal

Method

Preparing Agar Plates

63. Melt nutrient agar and allow to cool to $\sim 50^{\circ}\text{C}$ (remains liquid but not hot enough to harm bacteria).
64. Pour agar into sterile Petri dishes in a sterile environment (near Bunsen). Fill to $\sim 0.5\text{ cm}$ depth. Allow to set.

Preparing Plant Extracts

65. Crush/grind plant material (e.g. garlic, ginger, onion, mint leaves) with a pestle and mortar.
66. Add a small volume of distilled water or ethanol and filter to obtain the extract.
67. Soak filter paper discs (or use well-cut holes in agar) with plant extract.

Inoculating the Agar

68. Inoculate the agar plate by spreading a broth culture of the bacteria (e.g. *E. coli* or *Micrococcus luteus*) evenly across the surface using a sterile spreader or inoculation loop — this creates a 'lawn' of bacteria.
69. Using sterile forceps, place filter paper discs soaked in plant extracts onto the inoculated agar. Space them evenly.
70. Include control discs: one soaked in distilled water/ethanol (negative control), one soaked in a known antibiotic (positive control).

71. Replace lid, seal with adhesive tape (parafilm) to prevent contamination.
72. Label on the base and incubate inverted at 25–30 °C for 24–48 hours.

Recording Results

73. Measure the diameter of any clear zone (zone of inhibition) around each disc in mm using a ruler (measured through the lid — do NOT open plates after incubation).
74. Calculate area of inhibition: $A = \pi \times (d/2)^2$ for comparison.
75. Record and compare across plant extracts and controls.

Control Type	Expected Result
Water disc (negative control)	No zone of inhibition — confirms inhibition is from plant extract
Known antibiotic (positive control)	Clear zone of inhibition — confirms assay is working
Plant extract disc	Zone of inhibition if antimicrobial activity present

SAFETY & DISPOSAL
• All bacteria cultures: treat as potential pathogens — gloves, lab coat, eye protection
• After incubation: DO NOT open plates; measure through lid
• Autoclave or soak in disinfectant (e.g. 1% Virkon) before disposal
• Wash hands thoroughly after practical
• Do NOT use pathogenic organisms; use safe strains approved for school use

CORE PRACTICAL 10 (IA2 Unit 4)

Investigate the Effects of Light Intensity, Light Wavelength, Temperature and CO₂ Availability on the Rate of Photosynthesis Using a Suitable Aquatic Plant

SPECIFICATION REFERENCE

4.1 — Photosynthesis: light-dependent and light-independent reactions

Background

Aquatic plants (e.g. Elodea/Cabomba) release oxygen bubbles as a byproduct of the light-dependent reaction of photosynthesis. Counting bubbles or measuring O₂ concentration with a sensor gives a proxy for photosynthesis rate.

- Light intensity: inversely proportional to distance² (inverse square law)
- Wavelength: chlorophyll absorbs most in red (~680 nm) and blue (~430 nm) light
- Temperature: affects enzymes of the Calvin cycle (light-independent reactions)
- CO₂ : substrate for RuBisCO in Calvin cycle; increased CO₂ → increased rate (up to a limit)

Apparatus

- Elodea or Cabomba (freshwater aquatic plant)
- Beaker of water + sodium bicarbonate solution (0.5–1%) as CO₂ source
- Ruler, lamp (LED or tungsten), coloured filters (for wavelength investigation)
- Stopwatch, thermometer, O₂ sensor (optional)

Method: Counting Bubbles

76. Cut a fresh piece of Elodea (~10 cm) and place in a boiling tube with 0.5% sodium bicarbonate solution.
77. Illuminate with a lamp at a set distance (e.g. 10 cm). Allow 2 minutes to equilibrate.
78. Count bubbles produced per minute. Repeat 3 times; calculate mean.
79. Change the lamp distance (e.g. 10, 20, 30, 40, 50 cm) to alter light intensity. Record bubbles/min at each.
80. For wavelength: use the same distance but change coloured filters (red, orange, yellow, green, blue, clear).
81. For temperature: place the apparatus in water baths at different temperatures (10, 20, 30, 35 °C). Allow equilibration.
82. For CO₂ : vary NaHCO₃ concentration (0, 0.1%, 0.5%, 1%, 2%).

Inverse Square Law (Light Intensity)

LIGHT INTENSITY FORMULA

Light Intensity $\propto 1 / d^2$

where d = distance from light source to plant (cm)

Relative light intensity = $1/d^2$

e.g. At 10 cm: $I = 1/100 = 0.01$ units
At 20 cm: $I = 1/400 = 0.0025$ units
Plot rate of photosynthesis vs. $1/d^2$ to get a linear relationship

Controlling Variables

Controlled Variable	How to Control
CO ₂ concentration	Same concentration of sodium bicarbonate in all repeats
Temperature	Water bath; monitor with thermometer; ice if necessary
Plant species/size	Same plant, same length of cutting for each test
Background light	Darkened room or blackout box around apparatus

Limitations of Bubble Counting

- Bubbles vary in size — not a consistent measure of O₂ volume
- Better method: collect gas in inverted measuring cylinder (or use O₂ electrode)
- Photosynthetic pigments degrade over time — repeat quickly or keep plant healthy

SAFETY
• Electrical equipment near water: ensure no water splashes on lamp/wires
• Hot lamps: allow to cool; handle with cloth or stand
• Sodium bicarbonate: generally safe but wear goggles

CORE PRACTICAL 11 (IA2 Unit 4)

Carry Out a Study of the Ecology of a Habitat: Quadrats and Transects to Determine Distribution and Abundance; Measuring Abiotic Factors

SPECIFICATION REFERENCE

4.2 — Ecology: abiotic and biotic factors; sampling; biodiversity

Sampling Techniques

1. Random Sampling with Quadrats

A quadrat is a square frame (e.g. 0.5 m × 0.5 m or 1 m × 1 m) used to sample the plants or slow-moving animals in a defined area.

83. Use random number tables or a calculator to generate pairs of coordinates within the study area.
84. Place the quadrat at the coordinates and record: species present, number of individuals (density), percentage cover.
85. Repeat for a minimum of 10 quadrats (more is better).
86. Calculate mean density or percentage cover per species across all quadrats.

Percentage Cover vs. Frequency

Measure	Definition
Percentage cover	Estimated % of the quadrat area covered by each species
Frequency	Number of quadrats in which a species appears / total quadrats × 100%
Density	Number of individuals per unit area (per m ²)

2. Belt Transect (Systematic Sampling)

Used to show how species distribution changes across an environmental gradient (e.g. up a seashore, from a woodland edge into open field).

87. Lay a tape measure in a straight line across the habitat.
88. Place quadrats at regular intervals along the tape (e.g. every 2 m).
89. Record species composition at each quadrat.
90. Measure abiotic factors at each point.
91. Plot kite diagrams, bar charts, or describe zonation.

Measuring Abiotic Factors

Abiotic Factor	Measurement Method	Instrument
Light intensity	Lux meter placed at surface	Lux/light meter
Temperature	Air or soil temperature at each quadrat	Thermometer or data logger
Soil pH	Mix soil with distilled water; use pH probe or universal indicator	pH meter or colorimetric kit
Soil moisture	Soil moisture probe or gravimetric (weigh before/after drying)	Soil moisture meter

Wind speed	At each transect point	Anemometer
Humidity	Air humidity	Hygrometer
Oxygen concentration (water)	Dissolved O ₂ probe	O ₂ sensor/data logger

Statistical Analysis: Spearman's Rank Correlation

SPEARMAN'S RANK (rs)
Used to test whether two variables (e.g. light intensity and plant density) are correlated.
$rs = 1 - (6\sum d^2) / (n(n^2-1))$
where d = difference between ranks for each data pair
n = number of data pairs
rs = +1 → perfect positive correlation
rs = 0 → no correlation
rs = -1 → perfect negative correlation
Compare calculated rs to critical value at p = 0.05 to determine significance

Why Random Sampling?

- Avoids human bias in selection of sample locations
- Allows statistical analysis of the data
- More samples → more reliable estimate of the population

SAFETY
• Fieldwork: wear appropriate clothing (waterproof, sun protection)
• Be aware of hazards: uneven ground, tides (coastal), stinging plants
• Never work alone; inform school of fieldwork location
• Do not disturb habitats unnecessarily; follow countryside code

CORE PRACTICAL 12 (IA2 Unit 4)

Investigate the Effects of Temperature on the Development of Organisms (e.g. Seedling Growth Rate or Brine Shrimp Hatch Rates), Taking Into Account the Ethical Use of Organisms

SPECIFICATION REFERENCE

4.2 — Environmental factors and growth; enzymes in development

Background

Temperature affects all enzyme-catalysed reactions and therefore influences growth, development and metabolism. This practical explores how temperature affects biological processes such as seed germination, seedling growth, or brine shrimp hatching.

Option A: Seed Germination / Seedling Growth Rate

Method

92. Prepare sets of Petri dishes lined with wet filter paper. Place 10 seeds (e.g. mung beans, cress) in each.
93. Seal dishes with cling film to prevent drying out.
94. Incubate at different temperatures: e.g. 5, 10, 15, 20, 25, 30, 35 °C using incubators or water baths.
95. Record: time to germination (radicle emergence), number germinated (germination rate), length of shoot/root after 5 days.
96. Photograph or measure radicle/shoot length daily.
97. Calculate mean germination rate (%) and mean growth rate (mm/day).

Option B: Brine Shrimp (Artemia) Hatch Rates

Method

98. Add a pinch of Artemia cysts to 100 mL of artificial seawater (3% NaCl) in 250 mL beakers.
99. Place beakers at different temperatures (e.g. 10, 15, 20, 25, 30, 35 °C) and provide light.
100. After 24–48 hours, count nauplii (hatched shrimps) using a hand lens or microscope.
101. Calculate percentage hatch rate at each temperature.

Ethical Considerations

ETHICAL USE OF ORGANISMS

- Only use organisms in ways that minimise suffering (3Rs: Replace, Reduce, Refine)
- Brine shrimp: simple organisms — considered acceptable; dispose of humanely after experiment
- Seedlings: plant material is generally acceptable but avoid wasted material
- Do NOT use vertebrates for growth experiments without full ethics approval
- Record only what is necessary; minimise the number of organisms used

Expected Results & Analysis

- Hatch rate / germination rate increases up to an optimal temperature, then decreases
- This reflects the effect of temperature on enzyme activity controlling development

- Plot percentage hatch rate vs. temperature; identify optimum
- Q_{10} can be calculated: $Q_{10} = (\text{rate at } T+10^{\circ}\text{C}) / (\text{rate at } T^{\circ}\text{C})$ — usually ~2 for biological processes

SAFETY

- Incubators: do not exceed recommended temperatures (damage to samples or equipment)
- Brine shrimp culture: saline solution — avoid splashing in eyes
- General lab safety: wash hands after handling biological material

CORE PRACTICAL 13 (IA2 Unit 4)

Investigate the Rate of Growth of Microorganisms in a Liquid Culture, Taking Into Account the Safe and Ethical Use of Organisms

SPECIFICATION REFERENCE

4.3 — Microbiology: bacterial growth curve; aseptic technique; turbidity

Background

Bacteria in a liquid culture undergo characteristic growth phases: Lag, Exponential (Log), Stationary, and Decline (Death). Growth is measured by turbidity (cloudiness) using a colorimeter — more bacteria = more turbidity = higher absorbance.

Growth Phases

Phase	Description	What is happening
Lag phase	Initial slow growth; flat curve	Bacteria adapting to medium; synthesising enzymes
Exponential (log) phase	Rapid doubling; steep curve	Nutrients plentiful; maximum growth rate; constant generation time
Stationary phase	Growth rate = death rate; flat plateau	Nutrient depletion; accumulation of waste products
Decline (death) phase	Population decreases; curve drops	Nutrients exhausted; toxic waste accumulates; cells die

Method

102. Prepare sterile nutrient broth in a conical flask using aseptic technique (autoclaved broth).
103. Inoculate with a small measured volume of a culture of a suitable bacterium (e.g. *E. coli*, *Bacillus subtilis*, or yeast for safety). Use a sterile loop or pipette.
104. Incubate in a shaking incubator at 25 °C (school-safe temperature) to ensure oxygen supply and even distribution.
105. Every 30–60 minutes, remove a 1 cm³ sample using a sterile pipette (aseptic technique throughout).
106. Transfer sample to a cuvette and measure absorbance at 600 nm (turbidity) using a colorimeter.
107. Zero/blank the colorimeter with sterile broth before readings.
108. Record absorbance vs. time; plot to obtain a growth curve.

Calculating Generation Time

GENERATION TIME FORMULA

During exponential phase, bacteria double every 'generation time' (g)

$$n = (t_2 - t_1) / g \rightarrow g = (t_2 - t_1) / n$$

where n = number of generations, t = time
Alternative from growth data:
$g = 0.301 / \text{growth rate constant (k)}$
where $k = (\log_{10} N_2 - \log_{10} N_1) / (t_2 - t_1)$

Ethical Considerations

- Only use safe, non-pathogenic strains (Risk Group 1 organisms)
- Minimise volumes and cultures used
- All waste must be autoclaved or treated with disinfectant before disposal
- Plates/flasks must NEVER be opened after incubation outside the lab

SAFETY
• Treat all microorganism cultures as potential pathogens — gloves, lab coat, eye protection
• Work near a Bunsen burner; use aseptic technique throughout
• Autoclave all waste at 121 °C, 15 min, 15 psi before disposal
• Do NOT use organisms from clinical sources or unknown isolates

CORE PRACTICAL 14 (IA2 Unit 4)

Investigate the Effect of Different Antibiotics on Bacteria

SPECIFICATION REFERENCE

4.3 — Antimicrobials; antibiotic resistance; zones of inhibition

Background

Different antibiotics inhibit bacterial growth through different mechanisms (e.g. penicillin inhibits cell wall synthesis, streptomycin inhibits protein synthesis). Their effectiveness is measured by the size of the zone of inhibition around antibiotic-impregnated discs placed on a bacterial lawn.

Method

109. Use aseptic technique to make a uniform bacterial lawn on a nutrient agar plate (see CP9 for full aseptic technique guidance).
110. Using sterile forceps, place commercial antibiotic discs (Mast Rings or Oxoid Sensi-Discs) onto the agar surface. Space them ≥ 2 cm apart from edge and each other.
111. Include a control disc impregnated with distilled water (negative control).
112. Seal plate, label on base, incubate inverted at 25–30 °C for 24 hours.
113. WITHOUT opening the plates, measure the diameter of the clear zone of inhibition around each disc in mm.
114. Calculate the area of inhibition: $A = \pi(d/2)^2$
115. Compare areas across different antibiotics and/or different bacterial species.

Interpreting Zones of Inhibition

Zone Size	Interpretation
Large clear zone (>20 mm)	Bacterium is sensitive to this antibiotic
Small zone (8–14 mm)	Intermediate sensitivity — may require higher dose
No zone / tiny zone (<8 mm)	Bacterium is resistant to this antibiotic
No zone around control disc	Confirms any inhibition is due to antibiotic, not disc material

Antibiotic Resistance

- Resistance arises via mutation and natural selection: bacteria with resistance genes survive and reproduce
- Mechanisms: enzyme degradation (e.g. beta-lactamase breaks down penicillin), efflux pumps, altered target sites
- Multi-drug resistant organisms (MRSA, ESBL): major public health concern

SAFETY & DISPOSAL

- Only use commercially prepared, safe antibiotic discs
- Never open plates after incubation — measure through lid
- Autoclave or disinfect all plates after the experiment

- Wash hands thoroughly; wear gloves and lab coat throughout

- Incubate at ≤ 30 °C (prevent growth of human pathogens)

CORE PRACTICAL 15 (IA2 Unit 5)

Use an Artificial Hydrogen Carrier (Redox Indicator) to Investigate Respiration in Yeast

SPECIFICATION REFERENCE

5.1 — Respiration: glycolysis, Krebs cycle, oxidative phosphorylation; NAD

Background

During respiration, NAD^+ and FAD are reduced to NADH and FADH_2 as hydrogen atoms (electrons + protons) are transferred along metabolic pathways. A redox indicator (e.g. DCPIP or methylene blue) acts as an artificial hydrogen carrier — it is decolourised (reduced) by hydrogen atoms released during respiration.

- DCPIP: blue (oxidised) → colourless (reduced)
- Methylene blue: blue (oxidised) → colourless (reduced)

Faster decolourisation indicates a higher respiration rate in the yeast.

Variables

Variable Type	Example
Independent Variable	Temperature; substrate type (glucose vs. sucrose vs. fructose); pH
Dependent Variable	Time for DCPIP/methylene blue to decolourise (or absorbance change per minute)
Controlled Variables	Volume of yeast suspension, concentration of indicator, volume of substrate solution

Method (Temperature Variation)

116. Prepare a fresh yeast suspension (e.g. 1 g dried yeast in 20 cm³ warm water) and add glucose solution (2%) — allow to activate for 10–15 min.
117. Into each boiling tube/test tube, add: 2 cm³ DCPIP (or methylene blue) + a set volume (e.g. 1 cm³) of yeast suspension.
118. Add liquid paraffin on top to exclude air (prevents re-oxidation of indicator by O₂ — essential for anaerobic conditions).
119. Place tubes in water baths at different temperatures (e.g. 20, 30, 40, 50, 60 °C).
120. Start timing. Record time for complete decolourisation. Repeat ×3.
121. Calculate rate of respiration: Rate = 1 / time (in min⁻¹) or measure absorbance change.

Method (Substrate Variation)

122. Repeat above using the same temperature (e.g. 37 °C) but varying the substrate: glucose, sucrose, fructose, starch, water (control).
123. Starch is NOT expected to decolourise DCPIP (yeast lacks amylase to break it down) — this serves as a substrate control.

WHY USE LIQUID PARAFFIN?

- Oxygen re-oxidises DCPIP/methylene blue → it would turn blue again
- Layer of liquid paraffin seals the surface and maintains anaerobic conditions

- | |
|--|
| <ul style="list-style-type: none">• This ensures only yeast respiration (not atmospheric O₂) reduces the indicator |
| <ul style="list-style-type: none">• Without it, results are unreliable |

Expected Results

- Fastest decolourisation at ~30–40 °C (optimum for yeast enzymes)
- No decolourisation at very low temperatures (enzymes too slow) or high temperatures (enzymes denatured)
- Glucose and fructose decolourise DCPIP; sucrose may if yeast has invertase; starch may not

SAFETY
• DCPIP: irritant — gloves and eye protection
• Methylene blue: stains skin and clothing — wear gloves
• Yeast: generally safe; dispose of in normal waste
• Hot water baths: burn risk — use tongs

CORE PRACTICAL 16 (IA2 Unit 5)

Use a Simple Respirometer to Determine the Rate of Respiration and RQ of a Suitable Material (e.g. Germinating Seeds or Small Invertebrates)

SPECIFICATION REFERENCE

5.1 — Aerobic and anaerobic respiration; respiratory quotient (RQ); gas exchange

Background

A respirometer measures the change in gas volume during respiration. Organisms consume O_2 and release CO_2 . If CO_2 is absorbed (by soda lime/KOH), only the decrease in O_2 is measured. Comparing results with and without CO_2 absorber gives the RQ.

- Aerobic respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$; $RQ = 6CO_2 / 6O_2 = 1.0$ (carbohydrate)
- Fat oxidation: $RQ \approx 0.7$; Protein: $RQ \approx 0.9$

Apparatus Setup

A simple manometer respirometer uses capillary tubing containing coloured fluid (e.g. eosin or bromine water). The fluid moves along the tube as O_2 is consumed.

Tube Contents	Purpose	What is measured?
Tube A: organism + soda lime/KOH	Absorbs CO_2 ; only O_2 consumption measured	O_2 consumed only $\rightarrow x$ mm/min
Tube B: organism only (no absorber)	Both O_2 consumed and CO_2 released	Net change ($O_2 - CO_2$) $\rightarrow y$ mm/min
Control tube: dead/inert material + soda lime	Controls for pressure/temperature changes	Should show no movement

Method

124. Prepare the respirometer: place germinating seeds (pre-weighed) in Tube A with a small mesh basket of soda lime/KOH pellets. Set up Tube B identically but with cotton wool instead of soda lime.
125. Add a control tube (with similar mass of inert material + soda lime).
126. Seal all tubes with petroleum jelly/vaseline to make airtight. Attach to manometer (U-tube) or capillary tubing containing coloured fluid.
127. Allow 5 minutes to equilibrate (temperature stabilisation).
128. Open tap/valve and start timing. Record the position of the fluid meniscus every 2 minutes for 10–20 minutes.
129. Calculate the rate of movement ($mm\ min^{-1}$) in Tube A and Tube B.

Calculating RQ

RQ CALCULATION
$RQ = CO_2\ released / O_2\ consumed$
From respirometer data:
$O_2\ consumed = \text{movement in Tube A (soda lime)} = x\ mm/min$

Net change = movement in Tube B (no soda lime) = y mm/min
CO ₂ released = x - y (if y is negative, CO ₂ release > O ₂ consumption)
OR: CO ₂ released = O ₂ consumed - net gas change
RQ = CO ₂ released / O ₂ consumed
RQ = 1.0 → using carbohydrates
RQ = 0.7 → using fats
RQ > 1.0 → anaerobic respiration occurring

Sources of Error

Error	Solution
Temperature changes affect gas volume (not just respiration)	Use thermostatically controlled water bath; include control tube
Air leaks in apparatus	Seal all joints with petroleum jelly; test by closing before adding organisms
Variation in organism activity/mass	Use equal masses; take multiple readings for mean
Soda lime not fully absorbing CO ₂	Ensure fresh soda lime; add sufficient amount

SAFETY
• Soda lime (NaOH/Ca(OH) ₂): corrosive irritant — avoid skin/eye contact; wear gloves
• Organisms (small invertebrates): treat humanely (3Rs); return after experiment
• Petroleum jelly: generally safe

CORE PRACTICAL 17 (IA2 Unit 5)

Investigate the Effects of Exercise on Tidal Volume, Breathing Rate, Respiratory Minute Ventilation, and Oxygen Consumption Using Data from Spirometer Traces

SPECIFICATION REFERENCE

5.2 — Gas exchange and breathing; spirometry; effects of exercise

Background

A spirometer records lung volume changes over time on a kymograph (paper) or electronic trace. During exercise, breathing rate and depth (tidal volume) increase to supply more O_2 and remove more CO_2 . This is mostly a data interpretation practical — direct student spirometry is often replaced by analysis of provided traces.

Spirometer Measurements — Key Terms

Term	Definition	Typical Value (adult)
Tidal Volume (TV)	Volume of air in one normal breath (in or out)	~0.5 dm ³ (500 cm ³)
Breathing Rate (BR)	Number of breaths per minute	~15 breaths/min (rest)
Respiratory Minute Volume (RMV)	Total volume of air breathed per minute = TV × BR	~7.5 dm ³ /min (rest)
Vital Capacity (VC)	Maximum air exhaled after maximum inhalation	~4.5–5 dm ³
Residual Volume (RV)	Air remaining in lungs after maximum exhalation	~1.2 dm ³
Total Lung Capacity	VC + RV	~6 dm ³
Inspiratory Reserve Vol.	Extra air that can be inhaled above TV	~2.5 dm ³
Expiratory Reserve Vol.	Extra air that can be exhaled below TV	~1.5 dm ³

Reading a Spirometer Trace

- X-axis: time (seconds or minutes)
- Y-axis: lung volume (dm³ or cm³)
- Normal breathing: regular oscillations; peaks (inhale) and troughs (exhale)
- Tidal Volume: height of one complete oscillation
- Breathing Rate: count complete cycles per minute from the trace
- $RMV = TV \times BR$
- The trace baseline drifts DOWNWARD over time (soda lime in spirometer absorbs CO_2 ; O_2 being consumed)

Calculating O_2 Consumption from a Trace

O ₂ CONSUMPTION CALCULATION
The downward drift in the baseline of the trace represents O ₂ consumed.
O ₂ consumption rate = (drop in baseline volume over time) / time
Example: baseline drops by 0.6 dm ³ over 3 minutes
→ O ₂ consumption = 0.6 / 3 = 0.2 dm ³ /min
Note: This only works if the spirometer contains a CO ₂ absorber (soda lime)

Effects of Exercise on Spirometer Traces

Parameter	Effect of Exercise vs. Rest
Tidal Volume	Increases (deeper breaths)
Breathing Rate	Increases (more breaths per minute)
RMV	Increases significantly (TV × BR both rise)
O ₂ consumption	Increases (faster baseline drop in trace)
CO ₂ production	Increases (proportionally to O ₂ consumption)

Limitations of Spirometry

- Spirometers measure volume change, not actual lung volumes; Residual Volume cannot be measured
- Subject variation (age, sex, fitness, height) affects results
- Mouthpiece and nose clip can feel uncomfortable → affects breathing naturally

SAFETY
• If using actual spirometers: use disposable mouthpieces (hygiene)
• Do not use spirometers on students with asthma, COPD, or other respiratory conditions
• Ensure soda lime is fresh and not caustic to touch
• Monitor student health during exercise protocol

CORE PRACTICAL 18 (IA2 Unit 5)

Investigate the Production of Amylase in Germinating Cereal Grains

SPECIFICATION REFERENCE

5.3 — Gene expression; enzyme production; starch hydrolysis in germination

Background

During germination, cereal grains (e.g. barley) produce amylase enzymes in the aleurone layer in response to gibberellin (GA) released by the embryo. Amylase breaks down starch in the endosperm into maltose and glucose to fuel growth. This is a classic example of gene expression being controlled by a hormone.

- GA diffuses from embryo → aleurone layer
- GA activates genes coding for α -amylase
- α -Amylase is secreted into the endosperm, hydrolyses starch

Method

130. Prepare starch agar plates (nutrient agar + 1% starch solution poured into Petri dishes and allowed to set).
131. Isolate the embryo and aleurone layer from barley seeds by dissecting them carefully.
132. Alternatively: homogenise germinating barley grains at 0, 1, 2, 3, 4, 5 days of germination. Filter to obtain extract.
133. Cut wells into the starch agar using a cork borer (or use filter paper discs soaked in extract).
134. Add equal volumes of extract (or sections of grain) into the wells. Incubate plates at 25 °C for 24 hours.
135. Flood the plate with iodine solution. The clear halos around the wells show where starch has been digested by amylase.
136. Measure the diameter of the clear halo (zone of starch hydrolysis) in mm. Compare across days of germination.

Variables

Variable Type	Detail
Independent	Days of germination (0, 1, 2, 3, 4, 5 days) OR GA concentration added
Dependent	Diameter of clear halo on starch agar plate (mm)
Controlled	Temperature, starch concentration, incubation time, volume of extract applied

Gibberellin Experiment Extension

137. Compare non-germinated grains (halved to expose endosperm, embryo present) vs. grains with embryo removed vs. grains with embryo removed but treated with exogenous GA solution.
138. Without embryo: no GA → no amylase → no halo. With exogenous GA: amylase produced → halo appears.

This demonstrates that GA from the embryo controls amylase production.

Expected Results

- Halo diameter increases with days of germination (more amylase produced over time)

- Grains with embryo removed and NO GA: no halo (no amylase)
- Grains with embryo removed but GA added: halo present (GA triggers amylase production)

SAFETY
• Iodine solution: irritant, stains skin — wear gloves
• Scalpels for dissection: sharp; always cut away from body; use cutting mats
• Starch agar is a microbial growth medium: treat as potential contamination risk

MATHEMATICS & LAB SKILLS FOR IAL BIOLOGY

1. Calculating Means, Ranges and Standard Deviation

MEAN, RANGE & STANDARD DEVIATION
Mean ($\bar{x}$) = $\Sigma x / n$
Range = Maximum value – Minimum value
Standard Deviation (s) = $\sqrt{[\Sigma(x - \bar{x})^2 / (n - 1)]}$
Coefficient of Variation (%) = $(s / \bar{x}) \times 100$
Used to indicate spread/reliability of data; large SD = more variable data

2. Dilutions and Concentration Calculations

DILUTION FORMULA
$C_1 V_1 = C_2 V_2$
where C_1 = initial concentration, V_1 = initial volume
C_2 = final concentration, V_2 = final volume
Example: 2% stock glucose → make 1%:
$C_1 V_1 = C_2 V_2 \rightarrow 2 \times V_1 = 1 \times 10 \rightarrow V_1 = 5 \text{ cm}^3 \text{ stock} + 5 \text{ cm}^3 \text{ water}$
Serial dilution: each tube is diluted from the previous one

3. Magnification and Actual Size

MICROSCOPY CALCULATIONS
Magnification = Image size / Actual size
Actual size = Image size / Magnification
Image size = Actual size × Magnification
Always use SAME units for both measurements!
1 mm = 1000 μm ; 1 μm = 1000 nm; 1 mm = 10^{-3} m

4. Rate Calculations

Rate = Change in quantity / Time. E.g. Bubble rate = bubbles per minute. Enzyme rate = absorbance change per minute ($\Delta\text{AU}/\text{min}$). Always state units.

5. Percentages and Percentage Change

PERCENTAGE CALCULATIONS
Percentage = (value / total) $\times$ 100
Percentage change = ((new – original) / original) $\times$ 100
Percentage difference = (A – B / mean) $\times$ 100
Example: Mitotic index = (cells in mitosis / total cells) $\times$ 100

6. Graphs and Best-Fit Lines

- Always: label axes with quantity AND units (e.g. 'Absorbance (AU)')
- Use suitable scale; plot all data points; draw best-fit curve (not dot-to-dot)
- Gradient = $\Delta y / \Delta x$ (calculate from best-fit line, not individual points)
- Use tangent at a point for rate on a curve

7. Colorimetry

COLORIMETRY TECHNIQUE
Principle: measure how much light of a specific wavelength is absorbed by a solution
Filter selection: use a filter of COMPLEMENTARY colour to the solution
→ Red/orange solution: use blue/green filter (550 nm range)
→ Blue/purple solution: use red filter (600–650 nm range)
→ Clear solution: use UV or trial different filters
Calibration: blank the colorimeter with the solvent (distilled water or buffer)
Read absorbance: Beer-Lambert law — $A \propto \text{concentration}$ (in dilute solutions)
Create calibration curve: plot known concentrations vs absorbance → read unknowns from it

8. Respiratory Quotient (RQ)

RQ CALCULATIONS
RQ = Volume CO_2 produced / Volume O_2 consumed
Carbohydrates: RQ = 1.0

Fats:	RQ $\approx$ 0.7
Proteins:	RQ $\approx$ 0.9
Anaerobic:	RQ > 1.0 (CO ₂ produced without O ₂)
Mixed diet/metabolism:	intermediate values

9. Spearman's Rank Correlation

Already covered in CP11. Use when: both variables are continuous; data does not need to be normally distributed; sample size ≥ 7 pairs. Compare r_s to critical value from table at $p = 0.05$.

10. Inverse Square Law (Light Intensity)

INVERSE SQUARE LAW
$I \propto 1/d^2$
Relative light intensity at distance $d = 1/d^2$
Plot rate of photosynthesis vs. $1/d^2$ to confirm proportional relationship
Doubling distance $\rightarrow 4\times$ reduction in intensity

11. Chi-Squared Test

CHI-SQUARED (χ^2) TEST
Used to test whether observed results differ significantly from expected (e.g. genetics ratios)
$\chi^2 = \sum (O - E)^2 / E$
where O = observed frequency, E = expected frequency
Compare χ^2 to critical value at appropriate degrees of freedom ($df = n - 1$) and $p = 0.05$
If $\chi^2 >$ critical value: reject null hypothesis (significant difference)

12. Zones of Inhibition (Antibiotic Assays)

ZONE OF INHIBITION AREA
Measure diameter (d) of clear zone in mm (through lid of plate)
Area = $\pi \times (d/2)^2 = \pi \times r^2$

Comparing area gives a fairer comparison than just diameter

13. Q_{10} Temperature Coefficient

Q_{10} CALCULATION
$Q_{10} = (\text{Rate at } T + 10^\circ\text{C}) / (\text{Rate at } T)$
$Q_{10} \approx 2$ for most biological reactions (rate doubles per 10°C rise)
Q_{10} close to 1 = temperature-insensitive reaction
Example: Rate at 20°C = 5 mm/min; Rate at 30°C = 10 mm/min
$Q_{10} = 10/5 = 2.0$

